

AI/ML Reference Notes

Machine Learning

Machine learning enables computers to learn patterns from data and use those patterns to make predictions or decisions. A typical machine learning workflow includes collecting data, preprocessing it, selecting features, training a model, evaluating the model, and deploying the model.

Supervised Learning

Supervised learning uses labeled training data. Classification is used when the target consists of categories, while regression is used when the target is a continuous numerical value. Examples include spam classification and house-price prediction.

Overfitting

Overfitting occurs when a model learns the training data too closely, including patterns that do not generalize to new data. An overfit model can have excellent training performance but weaker validation or test performance.

Preventing Overfitting

Several techniques can reduce overfitting. Cross-validation provides a more reliable estimate of generalization performance. Regularization adds a penalty for model complexity. Other approaches include reducing unnecessary features, limiting model complexity, and collecting more representative training data.

Model Evaluation

Classification models can be evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. A confusion matrix shows the numbers of correct and incorrect predictions for each class. Regression models can be evaluated using metrics such as mean absolute error, mean squared error, and R-squared.

Feature Scaling

Feature scaling transforms numerical features to comparable ranges. Standardization commonly transforms a feature so that it has approximately zero mean and unit variance. Scaling is particularly useful for algorithms based on distances or feature magnitudes.

Decision Trees and Random Forests

A decision tree makes predictions by recursively splitting data according to feature conditions. Random forests combine predictions from multiple decision trees and can provide better generalization than a single

tree.

Support Vector Machines

Support Vector Machines, or SVMs, find decision boundaries that separate classes. The kernel method allows SVMs to model nonlinear relationships by mapping data into a different feature space.

Neural Networks

Neural networks contain interconnected layers of computational units called neurons. During training, the network adjusts its parameters to reduce prediction error. Deep neural networks contain multiple hidden layers and can learn complex representations.